

Assessment of the influence of vegetarian diet on the occurrence of erosive and abrasive cavities in hard tooth tissues.



INTRODUCTION: The aim of the study was to determine the potential relation between vegetarian diet and tooth erosion and abrasion. MATERIAL/METHODS: The examination included 46 vegetarians and the same number in the control group. Clinical research was carried out in order to detect the presence of abrasive and erosive changes and the level of hygiene in oral cavities. The questionnaire survey concerned dietary and hygienic habits. Statistical analysis of the data was conducted with Chi-square test and Mann-Whitney U test. The relations between following a vegetarian diet and the occurrence of non-carious cavities was tested with models of logistic regression. RESULTS: Tooth erosion was present among 39.1% of vegetarians and 23.9% of controls, while abrasion appeared among 26.1% and 10.9%, respectively, and the differences were statistically insignificant. The distribution of the changes was similar in both groups. Among vegetarians, significantly more frequent consumption of sour products (predominantly raw vegetables and fruit and tomatoes) was observed. The level of oral hygiene and hygienic habits were similar in both groups. The analysis of statistical regression did not reveal any relations between following a vegetarian diet and the occurrence of tooth erosion and abrasion. DISCUSSION: The results did not reveal any direct influence of vegetarian diet on the occurrence of erosive and abrasive changes. However, in the vegetarian group, more frequent consumption of some sour products and more commonly used horizontal brushing method were observed, with a slightly higher occurrence of non-carious cavities. Further research is required to obtain unambiguous conclusions.